An Explainable Multi-Stage Decision-Support Pipeline for Chest X-Ray Classification, Localization, and Structured Reporting

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Abstract—Chest X-ray review is a structured clinical workflow: identify findings, localize supporting evidence, and synthesize those observations into an actionable impression. Most automated chest X-ray systems stop at image-level classification and output a probability vector that is difficult to audit in clinical use. This work closes that gap with a three-stage pipeline on a 15-label public subset of the VinBigData chest X-ray dataset. The first stage performs multi-label image classification, where transfer learning is dominant and a Swin Transformer reaches 0.9665 macro AUC-ROC, while a lightweight 1.31M-parameter CNN trained from scratch reaches 0.9314. The second stage provides spatial evidence through YOLOv8m bounding-box localization at 0.3666 mAP@0.5. The third stage is an agentic reporting layer that consumes classifier probabilities and detector outputs to produce a structured summary with supported findings, uncertain findings, and explicit review guidance. On a 300-case evaluation, the integrated workflow reaches 0.750 exact-match accuracy and 0.9560 macro accuracy. The core contribution is a modular, auditable decision-support workflow in which each stage remains visible, replaceable, and operationally useful.

Index Terms—Chest X-ray, VinBigData, multi-label classification, Swin Transformer, YOLOv8, explainable AI, clinical decision support, structured reporting.

I. Introduction

Chest radiography accounts for roughly 40% of all diagnostic imaging worldwide, and that volume makes it a natural target for machine learning assistance. Clinical value, however, comes from fitting the radiologist's workflow, not from classification accuracy in isolation. A radiologist reading a chest X-ray screens for abnormalities, localizes the anatomical basis for each finding, judges confidence in the evidence, and composes a structured impression that another clinician can act on. Most existing automated systems address only image-level classification and stop at a probability vector that is opaque to the downstream user.

This gap between model output and clinical usability motivates the present work. Rather than optimizing a single classification metric, the project constructs a multi-stage pipeline that mirrors the radiologist's cognitive workflow:

1) Classification: determining which of 15 abnormality labels are present.
2) Localization: providing bounding-box evidence that shows where in the image the findings are supported.
3) Structured reporting: synthesizing the classification and localization outputs into a concise, auditable summary that distinguishes confident findings from uncertain ones and flags classifier-detector disagreement.

The system is built as a first-pass clinical assistant: it organizes the evidence so a clinician can review it faster, with direct visibility into what the model found, where it found it, and how confident each component is. Each pipeline stage is independently replaceable, so the workflow improves cleanly as better classifiers, detectors, or report generators become available.

The work makes three practical contributions: a reproducible classifier benchmark showing the value of transfer learning on the public VinBigData subset while retaining a lightweight from-scratch alternative for low-compute settings; a YOLOv8m localization branch that provides spatial supporting evidence; and an agentic reporting layer that merges classifier and detector outputs into a structured decision-support summary.

II. Related Work

Prior work on chest X-ray AI has addressed classification, localization, and report generation largely as separate research threads. CheXNet showed that a DenseNet trained on ChestX-ray14 could match radiologist-level sensitivity for pneumonia detection, establishing deep transfer learning as the dominant approach for image-level classification [2]. Pham et al. extended this to the VinDr-CXR dataset by pairing multi-label classification with lesion localization and demonstrating that the combined output improved inter-reader agreement, providing early evidence that localization support has direct clinical utility [3].

On the report-generation side, approaches have ranged from fine-grained label conditioning [9] and contrastive attention [10] to variational topic inference [11]. More recently, multimodal LLMs such as CXR-LLaVA [12] and GPT-4-based impression generators [13] have shown that

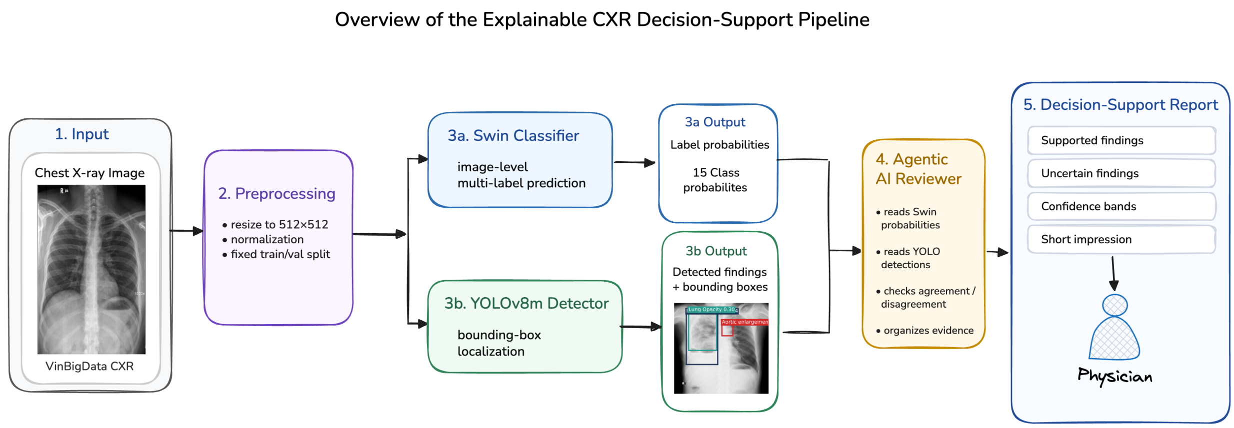

Fig. 1. Overview of the three-stage pipeline: chest X-ray input, Swin-based multi-label classification, YOLOv8 localization support, and agentic structured report generation.

large vision-language models can produce readable chest X-ray summaries. However, these systems are typically trained end-to-end on large report corpora (e.g., MIMIC-CXR) and treat the report as a single monolithic output, making it difficult to attribute a specific sentence in the generated text to a specific visual finding or model decision.

The present work targets a different design point. Instead of training a single end-to-end model, it assembles a modular pipeline from a bounded VinBigData subset with image-level labels and localization annotations, keeping the classifier, detector, and reporting stages explicitly separable. This modularity is the key architectural distinction: each stage's output is visible and auditable, so a clinician reviewing the final summary can trace any stated finding back to a classifier probability, a detector bounding box, or a disagreement between the two. The benefit is tangible: transparency and component-level replaceability are built into the workflow.

III. Dataset and Task Formulation

The pipeline operates on the publicly available Kaggle release of the VinBigData chest X-ray dataset [1], using the 15-label setup exposed by that competition. This public release is a truncated view of the broader original labeling space, yet it remains a strong fit for the pipeline because it contains localizable findings, diffuse findings, and the global No finding category in one benchmark.

The retained labels span a clinically meaningful range: Aortic enlargement, Atelectasis, Calcification, Cardiomegaly, Consolidation, ILD (Interstitial Lung Disease), Infiltration, Lung Opacity, Nodule/Mass, Other lesion, Pleural effusion, Pleural thickening, Pneumothorax, Pulmonary fibrosis, and No finding. Because a single radiograph may present multiple concurrent abnormalities, for example cardiomegaly alongside pleural effusion, the image-level task is formulated as multi-label classification rather than single-label assignment.

The localization task uses the raw VinBigData bounding-box annotations to train a YOLOv8m detector. These annotations are central to the pipeline: the public release provides boxes for abnormal findings, which allows image-level labels and spatial labels to remain aligned throughout the workflow. The classification and localization tasks are evaluated separately because they answer different clinical questions. Classification identifies what abnormalities are present anywhere in the image; localization identifies where the supporting evidence lies. The downstream reporting stage then uses agreement or disagreement between the two streams as an explicit confidence signal.

From an exploratory data analysis standpoint, the public labeled portion provides 15,000 studies, and this work uses a fixed split of 12,000 training images and 3,000 validation images. That split is reused across all experiments to ensure that performance differences reflect model choices rather than data sampling. The dataset is also visibly imbalanced: No finding is common, several chronic or diffuse abnormalities are relatively rare, and the frequency distribution is long-tailed. This is precisely why the dataset is useful for evaluating an auditable pipeline rather than a single closed classifier.

IV. Methods

A. Image-Level Classification

The classification stage answers the first clinical question: which abnormalities are present? All models were trained under a unified multi-label setup with input chest X-rays resized to 512×512, a fixed data split, and binary cross-entropy with logits as the loss function:

$$\begin{aligned}\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{NC}\sum_{i=1}^{N}\sum_{c=1}^{C}\Big[&y_{ic}\log\sigma(z_{ic}) \\ &+ (1-y_{ic})\log(1-\sigma(z_{ic}))\Big],\end{aligned} \tag{1}$$

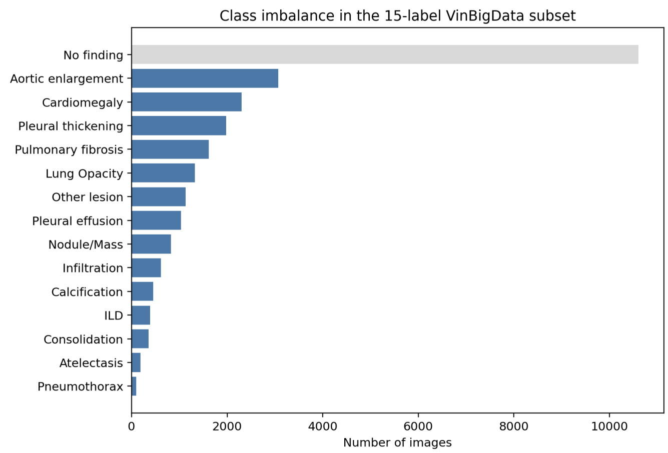

Fig. 2. Label frequency in the retained 15-label VinBigData subset. The dataset is strongly imbalanced, with No finding dominating and several abnormalities occurring rarely.

where N is the batch size, $C = 15$ is the number of labels, y_{ic} is the binary ground-truth label, and z_{ic} is the model logit for class c.

Five architectures were compared to understand the practical trade-off between transfer learning, model complexity, and classification performance in this clinical setting:

- efficientnet_b3: an ImageNet-pretrained EfficientNet-B3 [7] with a task-specific classification head,
- resnet50: an ImageNet-pretrained ResNet-50 [6] with a task-specific head,
- vit_base: an ImageNet-pretrained Vision Transformer [5] baseline, extended to 100 epochs given its slower convergence on this task,
- swin_base: an ImageNet-pretrained hierarchical Swin Transformer [4], whose shifted-window attention provides multi-scale feature extraction well suited to the varying spatial scales of chest X-ray abnormalities, and
- simple_cnn: a 1.31M-parameter CNN trained from scratch, serving as a resource-constrained baseline that tests whether acceptable performance is achievable without pretrained weights or large architectures.

For transfer learning to be possible, each pretrained backbone required a small but important architectural adaptation: the original single-label classification head was removed, the final projection layer was replaced with a 15-logit output head, and the forward pass was kept in a fully multi-label setting with sigmoid activations applied only at evaluation time. In practice, this means the pretrained visual encoder is reused as a generic feature extractor while the terminal prediction layer is specialized for chest X-ray abnormalities. This comparison serves the pipeline design: the chosen classifier determines the quality of downstream reporting, so understanding how much performance depends on architecture size and pretraining is directly relevant to deployment decisions.

B. Localization with YOLOv8m

The localization stage answers the second clinical question: where is the supporting evidence? A probability score alone does not tell the clinician which region of the image drove the prediction. YOLOv8m [8] is trained on the VinBigData bounding-box annotations to produce spatial evidence for each detected finding. The detector is evaluated using mAP@0.5 for detection quality, and its detections are additionally converted into image-level labels so that the detector's label-level recall can be compared against the classifier. This dual evaluation quantifies both localization precision and how much clinically relevant label signal is recoverable from bounding boxes alone.

In the pipeline, YOLO provides confirmatory or contradictory spatial evidence that the reporting stage references when generating the final summary.

C. Agentic Reporting Layer

The reporting stage addresses the third clinical question: how should the evidence be organized for review? In clinical practice, a radiologist does not hand a referring physician a list of probabilities; they produce a structured report with supported findings, uncertain observations, and an overall impression. The agentic reporting layer emulates this structure:

$$\begin{aligned} \text{CXR} &\rightarrow \text{classifier} \rightarrow \text{detector} \\ &\rightarrow \text{agentic AI} \rightarrow \text{structured report.} \end{aligned} \tag{2}$$

The layer consumes five inputs: (1) the full classifier probability vector, (2) predicted labels thresholded at 0.5, (3) YOLO detections above a confidence threshold of 0.25 and their associated scores, (4) bounding-box coordinates, and (5) explicit classifier-detector agreement or disagreement flags. From these, it generates a report containing supported findings (where both classifier and detector agree), uncertain findings (where only one source provides evidence), a short clinical impression, review guidance highlighting areas of model disagreement, and a safety statement. Architecturally, this layer follows explicit agentic design patterns: structured intermediate state instead of free-form prompt stuffing, provider-agnostic gateway access, schema-validated JSON outputs, deterministic decoding for reproducibility, and bounded retries when a response fails validation. These patterns make the reporting layer inspectable and stable on top of medical-model outputs. In these experiments, the LLM endpoint is Claude Sonnet, accessed through the Carnegie Mellon University AI gateway; temperature is fixed at zero for fully deterministic outputs.

To bridge the gap between low-level VinBigData labels and the broader categories a clinician uses when composing an impression, the report also maps findings into derived global buckets: No acute abnormality, Cardiomediastinal abnormality, Pleural abnormality, and Chronic interstitial or fibrotic pattern. These are interpretive summaries intended to improve readability; they are not treated as additional supervised labels. This local-versus-global distinction is central to the report design: some findings are naturally supported by boxes, while broader summary statements are derived from the total evidence bundle rather than from a single localized detection.

V. Experimental Setup

All training scripts use fixed defaults to ensure reproducibility. Most transfer-learning classifiers were trained for 10 epochs, which was sufficient given their pretrained starting points, while the from-scratch CNN was trained for 100 epochs to reach convergence without pretrained initialization. A follow-up ViT-Base run was extended to 100 epochs; the best validation checkpoint is reported. All classification models were trained with a batch size of 32 and used AdamW optimization with a learning rate of 10^{-4}, weight decay of 10^{-4}, and cosine learning rate scheduling. No data augmentation was applied during classifier training; images were resized to 512×512 and normalized with ImageNet channel statistics. The primary classification metric is macro AUC-ROC. This metric is the right headline measure for the setting because it is threshold-independent, gives each class equal weight despite the long-tailed frequency distribution, and prevents dominant classes such as No finding from masking weak rare-class behavior. Macro F1 and exact-match style metrics remain useful once thresholded decisions are required.

The end-to-end agentic evaluation is conducted on a 300-case slice drawn from the 3,000-image validation split; this subset was chosen to keep API usage within practical limits while providing a representative sample for pipeline comparison. The same 300 cases are used across all systems compared in Tables IV and V.

The YOLO localization model was initialized from COCO-pretrained YOLOv8m weights and trained on 512×512 images with a batch size of 16, using early stopping with a patience of 10 over up to 50 epochs; the best validation checkpoint is reported. All experiments were run using Pittsburgh Supercomputing Center resources with NVIDIA V100 GPUs, and the repository is structured so that any classification or localization run can be reproduced from the cleaned layout.

VI. Results

A. Image-Level Classification

Tables I and II summarize the classifier comparison. Table I reports the best validation AUC-ROC logged during training; Table II reports a complete post-training evaluation pass, which accounts for the small differences between corresponding AUC values across the two tables. Transfer learning is clearly dominant on this dataset. Swin achieves the strongest performance at 0.9665 macro AUC-ROC, followed closely by EfficientNet-B3 (0.9564) and ResNet-50 (0.9550). The best ViT-Base run reaches 0.8712 macro AUC-ROC, but it remains behind the CNN and the stronger pretrained backbones, confirming that a vanilla ViT is a weaker fit for this dataset even with longer training.

The most important secondary result is the from-scratch CNN: with only 1.31M parameters, it reaches 0.9314 macro AUC-ROC. The classification stage therefore does not require a large transformer to be effective. In a resource-constrained deployment, the lightweight CNN can serve as the classifier while the localization and reporting stages remain unchanged. This is a direct strength of the modular pipeline design.

At the same time, Swin's 3.5-point advantage over the CNN shows that stronger pretraining and hierarchical multi-scale attention do yield meaningful gains when compute is available. For the remainder of the pipeline evaluation, Swin is used as the primary classifier.

Table III shows how the best Swin checkpoint performs at the disease level. The aggregate metrics are strong because most classes are genuinely strong, not because a few easy classes are hiding weak ones. The clearest weak spots are Other lesion and thresholded Pneumothorax, while focal and cardiomediastinal findings are consistently strong.

The per-class breakdown exposes three distinct error regimes. First, broad and heterogeneous categories such as Other lesion (0.9156 AUC, 0.6747 F1) and Calcification (0.9349 AUC, 0.6271 F1) are limited by label ambiguity and visual diversity; these classes would benefit from stronger regularization, heavier dropout in the classification head, and class-balanced or focal-style losses that reduce dominance by frequent patterns. Second, classes such as Atelectasis (0.9598 AUC, 0.6076 F1) show good ranking quality but weaker thresholded performance, which points more directly to threshold calibration, label-specific decision thresholds, and probability calibration rather than to backbone weakness. Third, Pneumothorax shows the largest AUC–F1 gap (0.9733 vs. 0.4981), which strongly suggests that the model separates positives well but the default threshold is poorly matched to the class prevalence. For that regime, per-class threshold tuning is the highest-value intervention. Additional gains for diffuse findings such as ILD, Infiltration, and Lung Opacity are more likely to come from targeted augmentation, higher-resolution crops, and loss reweighting than from simply increasing model size.

B. Localization Performance

The best YOLOv8m run reached 0.3666 mAP@0.5; a larger YOLOv8l variant did not materially improve this. This result matches the task: VinBigData bounding boxes include diffuse abnormalities like lung opacity and ILD, which do not present the tight spatial boundaries object detectors handle best. Detection mAP alone therefore does not capture the detector's full value in the

TABLE I
Image-Level Classification Results on the 15-Label VinBigData Subset

Model	Params	Epochs	Macro AUC-ROC
simple_cnn	1.31M	100	0.9314
efficientnet_b3	11.49M	10	0.9564
resnet50	24.56M	10	0.9550
vit_base	86.84M	100	0.8712
swin_base	87.28M	10	0.9665

TABLE II
Best-Checkpoint Classification Metrics on the 3,000-Image Validation Split

Model	Exact	Macro Acc	Macro AUC-ROC
simple_cnn	0.7383	0.9444	0.9280
efficientnet_b3	0.7440	0.9543	0.9504
resnet50	0.7397	0.9526	0.9453
vit_base	0.6793	0.9313	0.8712
swin_base	0.7573	0.9583	0.9618

pipeline. YOLO provides a second, spatially grounded evidence channel. When the classifier flags cardiomegaly and the detector draws a bounding box over the cardiac silhouette, the reporting layer states that the finding has both image-level and localized support. When only one source provides evidence, the report marks the finding as uncertain. This agreement-disagreement signal is the core way localization strengthens clinical explainability. The YOLO labels row in Table V quantifies the label-level signal recoverable from bounding boxes alone: YOLO detections are converted to image-level predictions and compared directly against the classifier on the same 300-case slice. Fig. 3 shows representative detection outputs alongside radiologist ground truth for a clean single-label case and a harder multi-label case where the diffuse Lung Opacity finding is missed by the detector.

C. End-to-End Workflow Evaluation

Tables IV and V evaluate the complete pipeline on the 300-case review slice. Because the agentic layer produces categorical label decisions rather than soft probability scores, AUC for all three systems in this comparison is computed on thresholded binary predictions, placing every system on equal footing; this is why the AUC values here are substantially lower than those in Table II, where soft scores are used. The integrated workflow achieves 0.750 exact-match accuracy and 0.9560 macro accuracy, exceeding both Swin alone and YOLO labels on these metrics. On macro AUC-ROC it reaches 0.7063, below Swin's 0.7481 and above YOLO labels at 0.6374. The reporting layer therefore strengthens exact-match and macro accuracy while the AUC gap reflects the inherent cost of converting continuous scores to categorical outputs.

The case-level breakdown is illuminating. The agentic layer improved over Swin on 16 cases and over YOLO on 33 cases, while underperforming Swin on 12 cases and

TABLE III
Per-Class Metrics for the Best Swin Checkpoint

Finding	AUC-ROC	F1
Aortic enlargement	0.9826	0.9043
Atelectasis	0.9598	0.6076
Calcification	0.9349	0.6271
Cardiomegaly	0.9838	0.9086
Consolidation	0.9758	0.6993
ILD	0.9598	0.7587
Infiltration	0.9616	0.7763
Lung Opacity	0.9501	0.7744
Nodule/Mass	0.9464	0.7160
Other lesion	0.9156	0.6747
Pleural effusion	0.9758	0.8726
Pleural thickening	0.9485	0.7838
Pneumothorax	0.9733	0.4981
Pulmonary fibrosis	0.9663	0.8260
No finding	0.9927	0.9572
Macro average	0.9618	0.7590

TABLE IV
Matched 300-Case Slice: Classifier Alone vs. Full Pipeline

System	Exact-Match	Macro Acc	Macro AUC-ROC
Swin alone	0.7400	0.9542	0.7481
Full pipeline	0.7500	0.9560	0.7063

YOLO on 9 cases. The gains occurred when one model was uncertain and the other provided supporting evidence, allowing the reporting layer to make a stronger final call. The shortfalls occurred when the layer inherited a confident but incorrect upstream prediction. This asymmetry defines the fusion-based reporting approach: it strengthens weak signals and organizes evidence effectively, while upstream model quality still sets the performance ceiling.

The reporting stage delivers its strongest contribution at the workflow level. Instead of a single label vector, the clinician receives a structured summary that states which findings are supported by both models, which are uncertain, and where the two sources disagree. That information is invisible in a raw probability output. Preliminary comparisons against alternative agentic prompting strategies, including MedGemma-style medical LLM prompting, pointed to the same conclusion: the grounded pipeline produced the most auditable outputs because every report section is tied back to explicit classifier scores, detector boxes, and agreement logic. Medically specialized language models can phrase findings naturally, but the structured evidence contract in this pipeline makes the output far easier to audit as a decision-support component.

D. Qualitative Reporting Scenarios

Representative outcomes from the 300-case evaluation set illustrate where the workflow is most useful in practice. Vanilla scenario (minimal conflict). In the simplest workflow setting, the classifier, detector, and ground truth all align on either No finding or a small number of focal abnormalities. In these cases the pipeline produces

TABLE V
300-Case Comparison of the End-to-End Decision-Support Workflow

System	Exact-Match	Macro Acc	Macro AUC-ROC
Swin classifier	0.7400	0.9542	0.7481
YOLO labels	0.7300	0.9464	0.6374
Agentic workflow	0.7500	0.9560	0.7063

a conservative summary without forcing the clinician to sift through 15 individual probability scores. This is the simplest but most common workflow scenario, and handling it efficiently matters because low-conflict cases dominate clinical volume.

Medium-conflict scenario. In moderately conflicted cases, the classifier and detector agree on the major focal abnormality but differ on one secondary or diffuse finding. The report is useful here because it does not collapse the disagreement into a single opaque output. Instead, it preserves the supported finding as "backed by both sources" while explicitly marking the weaker finding as uncertain. A reviewing clinician can immediately see which part of the case is stable and which part still requires visual scrutiny.

High-conflict scenario. In the hardest cases, multiple findings are present and the evidence channels diverge substantially, for example when the classifier captures diffuse disease patterns that the detector cannot localize cleanly, or when one branch overcalls a focal lesion. The reporting layer is most valuable in these high-conflict settings because it surfaces the disagreement explicitly instead of hiding it inside an aggregate metric. The final report can then distinguish findings with dual support from classifier-only or detector-only claims, which is precisely the behavior needed for a first-pass clinical review tool.

Five representative example reports covering all three conflict regimes, together with their ground-truth comparison outputs, are included in the reports/example_reports/ and reports/comparision_reports/ directories of the source code submission for direct reference.

VII. Discussion

The principal finding of this work is that the value of the pipeline lies in its workflow design, not in any single component alone. A radiologist reviewing a chest X-ray identifies findings, locates supporting evidence, weighs certainty, and composes an impression. The pipeline mirrors that cognitive structure, and its clinical utility comes from making each step explicit and auditable.

This has immediate practical implications. A clinician interacting with the pipeline sees three layers of information: which abnormalities the classifier predicts, with probabilities; where the detector places supporting bounding boxes; and a structured summary that distinguishes well-supported findings from uncertain ones. If the classifier and detector disagree on a finding, the report says so. If the classifier is confident but the detector found no spatial evidence, the report marks the finding as uncertain. This transparency is a clear operational advantage over a single probability vector.

The modularity of the pipeline also sharpens deployment choices. The transfer-learning results show that pretrained backbones are the strongest default choice when compute is available, especially Swin. The CNN baseline shows that the same architecture still works well with a 1.31M-parameter model. The pipeline therefore scales cleanly from high-compute to resource-constrained settings without architectural redesign.

The localization branch's 0.3666 detection mAP should be interpreted in the context of its role. It is a confirmatory evidence channel inside the report, not an isolated endpoint. Its value is strongest for focal abnormalities such as aortic enlargement, cardiomegaly, and nodules, where bounding boxes align directly with anatomical findings. For diffuse abnormalities like ILD or lung opacity, the detector is weaker, and the pipeline responds directly by reporting the finding as classifier-only.

The agentic reporting layer improves exact-match performance over Swin alone (0.750 vs. 0.740) and transforms raw model outputs into a document that matches the structured reporting format radiologists already use: supported findings, uncertain observations, an impression, and review guidance. The implementation also surfaces an important engineering consideration about LLM provider selection. The meaningful comparison is not simply "which LLM writes the best prose," but which provider operates reliably inside a constrained orchestration loop with schema validation, deterministic settings, and evidence-grounded prompts. The design patterns around the agent are therefore as important as the model endpoint itself.

VIII. Limitations

Several limitations define the next phase of the work. First, the study uses the public 15-label Kaggle view of VinBigData rather than the full original taxonomy, so evaluation does not yet cover every diagnostic nuance in the broader dataset. Second, the agentic reporting layer has not yet been validated against radiologist-authored free-text reports in a clinical reader study; the structured summaries are evaluated against ground-truth labels. Third, the reporting layer inherits upstream errors and therefore depends directly on classifier and detector quality. Fourth, the MedGemma and provider-level comparisons described here are not yet a fully standardized benchmark. Fifth, the analysis emphasizes aggregate metrics and does not yet include calibration studies, external validation on out-of-distribution data, or subgroup analyses across patient demographics.

These limitations define a clear path forward. The system already provides an explainable decision-support workflow with a reproducible code path. The next step is prospective reader studies, calibration analysis, and external validation across different acquisition protocols and patient populations.

IX. Conclusion

This work presents a multi-stage chest X-ray decision-support pipeline that closes the gap between model output and clinical usability. The pipeline mirrors the radiologist's workflow: classification, localization, and structured reporting, while keeping each stage modular and auditable. Transfer-learned classifiers, especially Swin, provide the strongest image-level performance; a lightweight CNN keeps the same workflow viable under tighter compute constraints; YOLO contributes spatial evidence that strengthens explainability for focal findings; and the agentic reporting layer converts raw model outputs into a structured summary that a clinician can review and act on. The result is a practical, evidence-grounded clinical assistant that replaces opaque probability outputs with a structured foundation for review.

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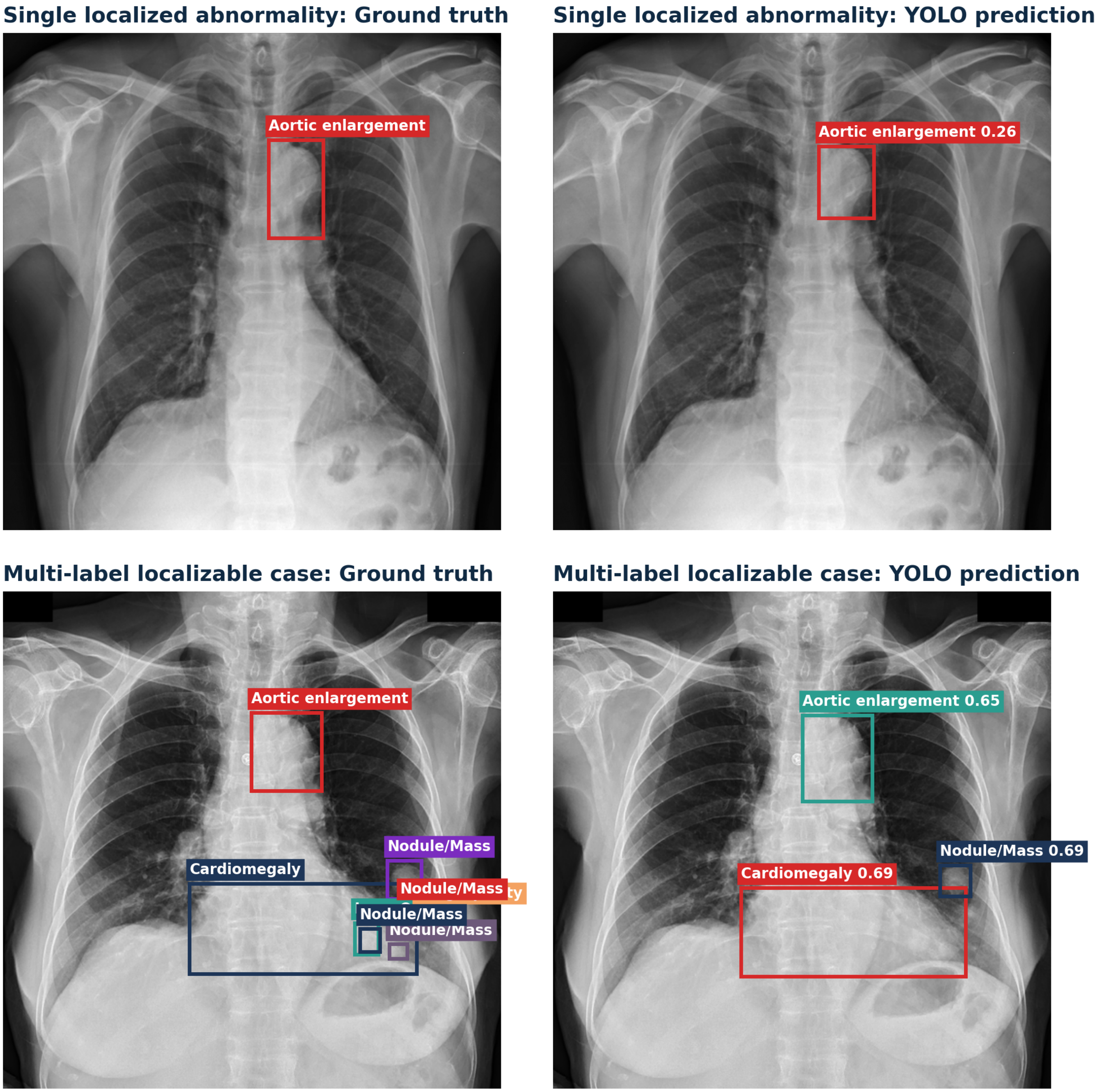

Fig. 3. Ground-truth-versus-YOLO comparison examples. In each row, the left panel shows merged radiologist ground-truth boxes and the right panel shows YOLO predictions with visible class labels. Top row: a single-label case where both the ground truth and YOLO indicate Aortic enlargement. Bottom row: a multi-label case with radiologist labels Aortic enlargement, Cardiomegaly, Lung Opacity, and Nodule/Mass; YOLO recovers the first three localizable patterns except the more diffuse Lung Opacity finding.

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